



VIDYAVARDHINI'S COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

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Experiment no.5

Aim : To understand the Kubernetes Cluster Architecture, install and deploy Kubernetes on Windows

Theory :

Kubernetes (K8s) is an open-source container orchestration platform developed by Google and now maintained by the Cloud Native Computing Foundation (CNCF). It automates the deployment, scaling, and management of containerized applications across machines.

Kubernetes is widely used for DevOps, microservices, and cloud-native applications due to its reliability, scalability, and portability.

2. Need for Kubernetes:

Modern applications are built using containers (like Docker) for consistency and isolation. However, managing multiple containers across many hosts manually is complex. Kubernetes simplifies this by:

- Automating container deployment and scaling
 - Handling load balancing
 - Providing self-healing capabilities (restarts failed containers automatically)
 - Offering service discovery and networking between containers
-

3. Kubernetes Cluster Architecture:

A Kubernetes cluster consists of two main types of nodes:

- Master Node (Control Plane)
 - Worker Nodes
-

A. Master Node (Control Plane):



The master node manages and controls the entire cluster. It makes global decisions about scheduling and monitors the cluster's state.

Key Components:

1. API Server:

- Acts as the front-end for the Kubernetes control plane.
- All communication (CLI, Dashboard, or REST calls) goes through it.

2. etcd:

- A distributed key-value store that stores all cluster configuration data (like node info, pod states, secrets, etc.).

3. Controller Manager:

- Monitors the cluster and ensures the desired state is maintained (e.g., if a pod crashes, it creates a new one).

4. Scheduler:

- Assigns newly created pods to appropriate worker nodes based on resource availability and constraints.
-

B. Worker Node:

Worker nodes are the machines where application containers actually run.

Key Components:

1. Kubelet:

- Agent running on each node.
- Ensures containers are running in a Pod as defined by the control plane.

2. Kube Proxy:

- Handles network communication and load balancing for pods and services.

3. Container Runtime:

- Responsible for running containers (e.g., Docker, containerd).

Each worker node hosts one or more Pods, which are the smallest deployable units in Kubernetes.



4. Kubernetes Objects and Concepts:

- **Pod:** Smallest deployable unit; can contain one or more containers.
- **ReplicaSet:** Ensures a specified number of identical pods are running.
- **Deployment:** Manages rolling updates and rollbacks for applications.
- **Service:** Exposes a set of pods as a network service.
- **Namespace:** Logical grouping of cluster resources.

Procedure:

Installing kubectl:

The screenshot shows the Kubernetes documentation website. The browser address bar displays <https://kubernetes.io/docs/tasks/tools/>. The page header includes the Kubernetes logo and navigation links: Documentation, Kubernetes Blog, Training, Partners, Community, Case Studies, Versions, and English. A banner for KubeCon + CloudNativeCon 2024 is visible. The left sidebar contains a search bar and a navigation menu with categories like Getting started, Concepts, Tasks, and Administer a Cluster. The main content area is titled 'Install Tools' and includes the sub-header 'Set up Kubernetes tools on your computer.' Below this, the 'kubectl' section is highlighted, describing it as the Kubernetes command-line tool. A right-hand sidebar offers options to edit the page, create a child page, create documentation, or print the entire page, along with a list of related tools: kubectl, kind, minikube, and kubeadm.



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Browser address bar: <https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/>

Navigation bar: **kubernetes** | Documentation | Kubernetes Blog | Training | Partners | Community | Case Studies | Versions | En

Hero section: **KubeCon + CloudNativeCon 2024**
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 - Install and Set Up kubectl on Windows**
 - Administer a Cluster

Kubernetes Documentation / Tasks / Install Tools / Install and Set Up kubectl on Windows

Install and Set Up kubectl on Windows

Before you begin

You must use a kubectl version that is within one minor version difference of your cluster. For example, a v1.31 client can communicate with v1.30, v1.31, and v1.32 control planes. Using the latest compatible version of kubectl helps avoid unforeseen issues.

Install kubectl on Windows

```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Kubernetes>curl.exe -LO "https://dl.k8s.io/release/v1.31.0/bin/windows/amd64/kubectl.exe"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 138 100 138 0 0 474 0 --:--:-- --:--:-- --:--:-- 479
100 55.2M 100 55.2M 0 0 1793k 0 0:00:31 0:00:31 --:--:-- 1820k

C:\Kubernetes>curl.exe -LO "https://dl.k8s.io/v1.31.0/bin/windows/amd64/kubectl.exe.sha256"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 138 100 138 0 0 429 0 --:--:-- --:--:-- --:--:-- 432
100 64 100 64 0 0 183 0 --:--:-- --:--:-- --:--:-- 183

C:\Kubernetes>CertUtil -hashfile kubectl.exe SHA256
SHA256 hash of kubectl.exe:
a618de26c86421a394de7041f9d0a87752dd4e555894d2278421cf12097fa531
CertUtil: -hashfile command completed successfully.

C:\Kubernetes>type kubectl.exe.sha256
a618de26c86421a394de7041f9d0a87752dd4e555894d2278421cf12097fa531
C:\Kubernetes>$(Get-FileHash -Algorithm SHA256 .\kubectl.exe).Hash -eq $(Get-Content .\kubectl.exe.sha256)
'$' is not recognized as an internal or external command,
operable program or batch file.

C:\Kubernetes>kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2

C:\Kubernetes>
```



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Installing minikube:

minikube start

minikube is local Kubernetes, focusing on making it easy to learn and develop for Kubernetes.

All you need is Docker (or similarly compatible) container or a Virtual Machine environment, and Kubernetes is a single command away: `minikube start`

What you'll need

- 2 CPUs or more
- 2GB of free memory
- 20GB of free disk space
- Internet connection
- Container or virtual machine manager, such as: Docker, QEMU, Hyperkit, Hyper-V, KVM, Parallels, Podman, VirtualBox, or VMware Fusion/Workstation

1 Installation

Click on the buttons that describe your target platform. For other architectures, see the release page for a complete list of

minikube

Click on the buttons that describe your target platform. For other architectures, see the release page for a complete list of minikube binaries.

Operating system:

Architecture:

Release type:

Installer type:

To install the latest minikube **stable** release on **x86-64 Windows** using **.exe download**:

1. Download and run the installer for the [latest release](#).
Or if using PowerShell, use this command:

```
type Directory -Force  
ikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
```

2. Add the `minikube.exe` binary to your `PATH`.
Make sure to run PowerShell as Administrator.

```
$oldPath = [Environment]::GetEnvironmentVariable('Path', [EnvironmentVariableTarget]::Machine)  
if ($oldPath.Split(';') -notcontains 'C:\minikube'){  
    [Environment]::SetEnvironmentVariable('Path', $('{};C:\minikube' -f $oldPath), [EnvironmentVariableTarget]::Machine)  
}
```



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```
PS C:\> New-Item -Path 'C:\' -Name 'minikube' -ItemType Directory -Force

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----         18-10-2024    05:04         minikube

PS C:\> Invoke-WebRequest -OutFile 'C:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
PS C:\> |
```

```
Administrator: Windows PowerShell

PS C:\> $oldPath = [Environment]::GetEnvironmentVariable('Path', [EnvironmentVariableTarget]::Machine)
>> if ($oldPath.Split(';') -notcontains 'C:\minikube'){
>>   [Environment]::SetEnvironmentVariable('Path', $('{0};C:\minikube' -f $oldPath), [EnvironmentVariableTarget]::Machine)
>> }
>>
PS C:\> _
```

```
C:\minikube>minikube start
* minikube v1.34.0 on Microsoft Windows 11 Home Single Language 10.0.22631.4317 Build 22631.4317
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.45 ...
* Updating the running docker "minikube" container ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Preparing Kubernetes v1.31.0 on Docker 27.2.0 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default

C:\minikube>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubecfg: Configured

C:\minikube>
```



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Installing Docker :

Check for System requirements:

The screenshot shows the Docker Docs website at <https://docs.docker.com/desktop/install/windows-install/>. The left sidebar contains a navigation menu with categories like Manuals, Docker Build, Docker Build Cloud, Docker Compose, Docker Desktop, and Install. The main content area is titled 'System requirements' and lists the following:

- WSL 2 backend, x86_64
- Hyper-V backend, x86_64
- WSL 2 backend, Arm (Beta)
- WSL version 1.1.3.0 or later.
- Windows 11 64-bit: Home or Pro version 21H2 or higher, or Enterprise or Education version 21H2 or higher.
- Windows 10 64-bit:
 - We recommend Home or Pro 22H2 (build 19045) or higher, or Enterprise or Education 22H2 (build 19045) or higher.
 - Minimum required is Home or Pro 21H2 (build 19044) or higher, or Enterprise or Education 21H2 (build 19044) or higher.
- Turn on the WSL 2 feature on Windows. For detailed instructions, refer to the [Microsoft documentation](#).
- The following hardware prerequisites are required to successfully run WSL 2 on Windows 10 or Windows 11:
 - 64-bit processor with [Second Level Address Translation \(SLAT\)](#).
 - 4GB system RAM

The screenshot shows the Docker Docs website at <https://docs.docker.com/desktop/install/windows-install/>. The left sidebar contains a navigation menu with categories like Manuals, Docker Build, Docker Build Cloud, Docker Compose, Docker Desktop, and Install. The main content area is titled 'Install Docker Desktop on Windows' and lists the following:

- Home / Manuals / Docker Desktop / Install / Windows
- Docker Desktop terms
- Commercial use of Docker Desktop in larger enterprises (more than 250 employees OR more than \$10 million USD in annual revenue) requires a [paid subscription](#).
- This page contains the download URL, information about system requirements, and instructions on how to install Docker Desktop for Windows.
- Docker Desktop for Windows - x86_64
- Docker Desktop for Windows - Arm (Beta)
- For checksums, see [Release notes](#)
- System requirements
- Tip

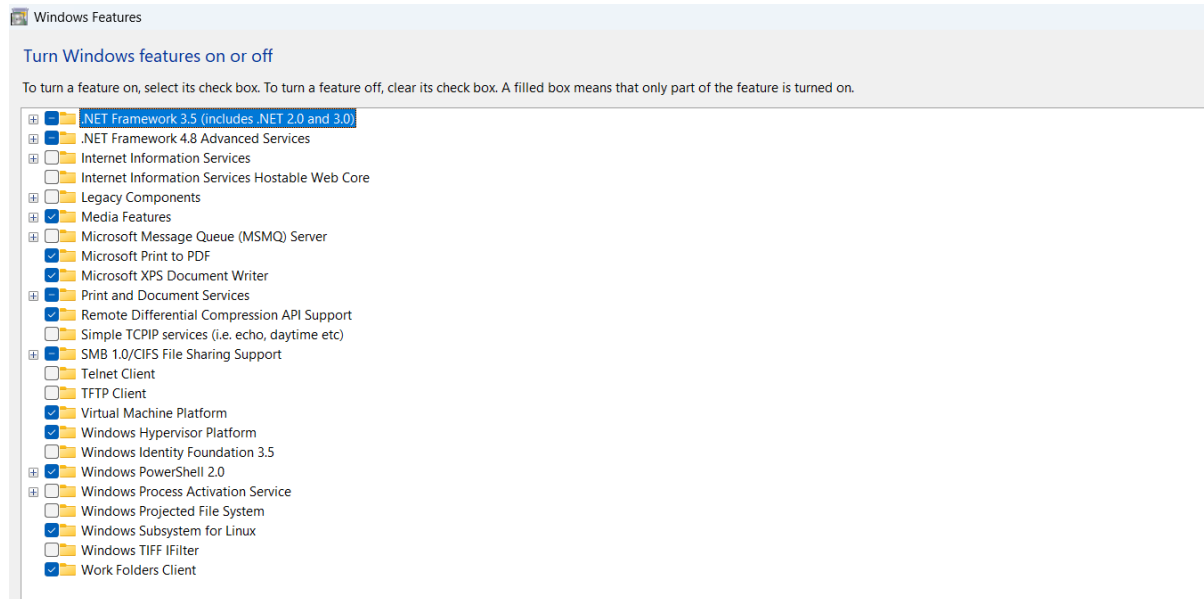


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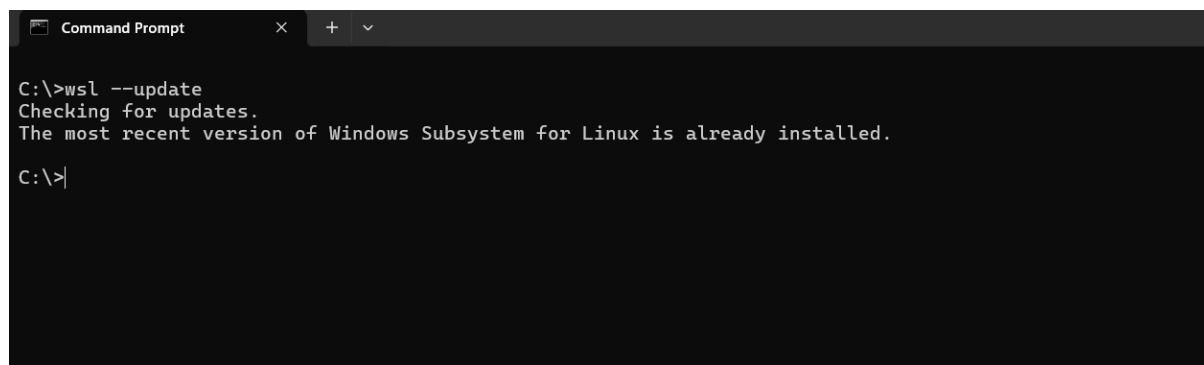
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Make sure you have selected :Virtual Machine Platform,Windows Hypervisor Platform and Windows Subsystem for Linux and restart your system once selected to run docker smoothly.



Command to update wsl:





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The screenshot shows the Docker Desktop interface. On the left is a sidebar with navigation options: Containers, Images, Volumes, Builds, Docker Scout, and Extensions. The main area is titled 'Containers' and displays system metrics: 'Container CPU usage' at 0.00% / 800% (8 CPUs available) and 'Container memory usage' at 0B / 3.67GB. A search bar and a toggle for 'Only show running containers' are present. Below is a table of containers:

Name	Image	Status	Port(s)	CPU (%)	Last started	Actions
minikube	k8s-minikube/kicbase:v0.0.45	Running	52271:22	0%	2 minutes ago	[Stop] [Refresh] [Delete]

Below the table, there are 'Walkthroughs' for 'Multi-container applications' (8 mins) and 'Containerize your application' (3 mins). The bottom status bar shows 'Engine running', system resources (RAM 3.84 GB, CPU 2.37%, Disk ~GB avail. of ~GB), and version 'v4.34.3'.

This screenshot shows the 'Logs' tab for the 'minikube' container. The breadcrumb navigation is 'Containers / minikube'. The container details show it is running with ID '0efa6d35141f' and image 'k8s-minikube/kicbase:none' (was k8s-minikube/kicbase:v0.0.45). The status is 'Running (4 minutes ago)' with control buttons for stop, play, refresh, and delete. The logs tab is selected, displaying a list of system boot logs with timestamps and status indicators (OK).

```
2024-10-18 06:46:53 Starting Docker Socket for the API...
2024-10-18 06:46:53 Starting Podman API Socket...
2024-10-18 06:46:53 [ OK ] Listening on Docker Socket for the API.
2024-10-18 06:46:53 [ OK ] Listening on Podman API Socket.
2024-10-18 06:46:53 [ OK ] Reached target Socket Units.
2024-10-18 06:46:53 [ OK ] Reached target Basic System.
2024-10-18 06:46:53 Starting containerd container runtime...
2024-10-18 06:46:53 Starting minikube automount...
2024-10-18 06:46:53 Starting Podman auto-update service...
2024-10-18 06:46:53 Starting Podman Start All _restart Policy Set To Always...
2024-10-18 06:46:53 Starting Podman API Service...
2024-10-18 06:46:53 Starting OpenBSD Secure Shell server...
2024-10-18 06:46:53 [ OK ] Started Podman API Service.
2024-10-18 06:46:53 [ OK ] Finished minikube automount.
2024-10-18 06:46:53 [ OK ] Started OpenBSD Secure Shell server.
2024-10-18 06:46:55 [ OK ] Started containerd container runtime.
2024-10-18 06:46:55 Starting Docker Application Container Engine...
2024-10-18 06:46:55 [ OK ] Finished Podman Start All _restart Policy Set To Always.
2024-10-18 06:46:55 [ OK ] Finished Podman auto-update service.
2024-10-18 06:46:57 [ OK ] Started Docker Application Container Engine.
2024-10-18 06:46:57 [ OK ] Reached target Multi-User System.
2024-10-18 06:46:57 [ OK ] Reached target Graphical Interface.
2024-10-18 06:46:57 Starting Record Runlevel Change in UTMP...
2024-10-18 06:46:57 [ OK ] Finished Record Runlevel Change in UTMP.
```

The bottom status bar shows system resources (RAM 3.85 GB, CPU 1.26%, Disk ~GB avail. of ~GB) and version 'v4.34.3'.



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```
C:\minikube>kubectl get po -A
NAMESPACE      NAME                                     READY   STATUS    RESTARTS   AGE
kube-system    coredns-6f6b679f8f-bq25z             1/1     Running   1 (18m ago) 23m
kube-system    etcd-minikube                         1/1     Running   1 (18m ago) 23m
kube-system    kube-apiserver-minikube               1/1     Running   1 (18m ago) 23m
kube-system    kube-controller-manager-minikube      1/1     Running   1 (18m ago) 23m
kube-system    kube-proxy-k4459                     1/1     Running   1 (18m ago) 23m
kube-system    kube-scheduler-minikube               1/1     Running   1 (18m ago) 23m
kube-system    storage-provisioner                   1/1     Running   3 (73s ago) 23m

C:\minikube>kubectl create deployment hello-minikube --image=kicbase/echo-server:1.0
deployment.apps/hello-minikube created

C:\minikube>kubectl expose deployment hello-minikube --type=NodePort --port=8080
service/hello-minikube exposed
```

```
C:\minikube>kubectl get services hello-minikube
NAME          TYPE       CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
hello-minikube NodePort    10.102.156.18 <none>       8080:32254/TCP  116s

C:\minikube>minikube service hello-minikube
|-----|
| NAMESPACE | NAME          | TARGET PORT | URL                  |
|-----|
| default   | hello-minikube | 8080        | http://192.168.49.2:32254 |
|-----|
* Starting tunnel for service hello-minikube.
|-----|
| NAMESPACE | NAME          | TARGET PORT | URL                  |
|-----|
| default   | hello-minikube |             | http://127.0.0.1:53604 |
|-----|
* Opening service default/hello-minikube in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



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```
127.0.0.1:53604 x +
http://127.0.0.1:53604
Classes Oral questions on C... Oracle Live SQL - S... (20) How To Make A... (10) 15: How to Inse... (13) 5: We Use Boxe... Call Of Duty: Black...
Request served by hello-minikube-7d48979fd6-mqng7
HTTP/1.1 GET /
Host: 127.0.0.1:53604
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: en-US,en;q=0.9,hi;q=0.8
Connection: keep-alive
Sec-Ch-Ua: "Google Chrome";v="129", "Not=A?Brand";v="8", "Chromium";v="129"
Sec-Ch-Ua-Mobile: ?0
Sec-Ch-Ua-Platform: "Windows"
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Sec-Fetch-User: ?1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36
```

Conclusion :

Kubernetes provides a powerful, automated, and scalable platform for managing containerized applications. By understanding its cluster architecture and deploying it on Windows, developers can experiment with real-world DevOps workflows, microservices management, and cloud-native development in a local environment.